

NETWORK TRAFFIC ANALYSIS REPORT

Instagram & Facebook Mobile Application Traffic

Captured via Wireshark | Android Device | 2025

✓ No Suspicious Activity Detected | Portfolio Project

1. Executive Summary

This report documents a personal cybersecurity project in which live network traffic generated by the Instagram and Facebook mobile applications was captured and analyzed using Wireshark. The objective was to examine protocol behavior, identify data types transmitted, understand communication patterns, and determine whether any anomalous or suspicious activity was present.

No malicious behavior, unauthorized data exfiltration, or suspicious connections were detected. All traffic conformed to expected patterns for modern social media applications, utilizing TLS 1.3 encryption and well-known infrastructure operated by Meta Platforms, Akamai, and Cloudflare CDN networks.

This project demonstrates practical skills in packet capture methodology, protocol analysis, traffic classification, and security assessment — core competencies for cybersecurity, SOC analyst, and network forensics roles.

2. Why Packet Analysis Matters in Cybersecurity

2.1 For Security Operations (Blue Team / SOC)

- Detecting Command & Control (C2) traffic from malware beaconing
- Identifying data exfiltration — unauthorized outbound data transfers
- Investigating lateral movement across internal networks
- Incident response: reconstructing attack timelines from PCAP evidence
- Validating that sensitive data is encrypted end-to-end

2.2 For Penetration Testing (Red Team)

- Intercepting unencrypted credentials, tokens, or API keys
- Identifying insecure protocols (HTTP, FTP, Telnet) still in use
- Testing for certificate pinning bypasses in mobile applications
- Mapping target infrastructure through passive DNS enumeration

2.3 For Digital Forensics

- Collecting legally admissible network evidence during incidents
- Reconstructing file transfers, browsing history, and communications
- Attributing activity to specific devices via MAC/IP correlation

2.4 Why Social Media App Analysis Is Specifically Valuable

Instagram and Facebook are among the most widely installed apps globally. Analyzing their traffic helps analysts recognize legitimate Meta/CDN traffic vs. spoofed/malicious traffic mimicking it, detect metadata leakage in encrypted sessions, and build baselines for behavioral anomaly detection.

3. Methodology

3.1 Environment Setup

Parameter	Details
Capture Tool	Wireshark (Latest Stable)
Device	Android Smartphone
Network Interface	Wi-Fi (802.11)
Capture File Format	.pcapng (Next Generation PCAP)
Target Applications	Instagram, Facebook (Meta Platforms)
Capture Filter	None (promiscuous mode — all traffic)

3.2 Capture Process

- Connected Android device and capture machine to the same Wi-Fi network
- Launched Wireshark and selected the appropriate network interface
- Started capture before opening Instagram and Facebook apps
- Performed standard interactions: feed scrolling, profile loading, stories
- Stopped capture and saved as .pcapng for structured analysis

3.3 Analysis Approach

- Protocol Hierarchy Statistics: identified distribution across all protocol layers
- Conversation tracking: mapped all IP endpoints contacted by the apps
- DNS query inspection: documented all domains resolved during the session
- TLS/SSL inspection: verified handshake parameters and certificate validity
- Security screening: checked for plaintext credentials and anomalous patterns

4. Types of Data Collected & Observed

4.1 Protocol Distribution

Protocol	Description & Role in Traffic
Ethernet (Layer 2)	Physical/data link framing for all packets on the local network
IPv4 / IPv6 (Layer 3)	Network addressing; IPv6 increasingly used by Meta infrastructure
TCP (Layer 4)	Reliable transport for all HTTPS/TLS streams

UDP (Layer 4)	Used by QUIC protocol and DNS queries
DNS	Resolves hostnames to IPs before connections are established
TLS 1.3 (Layer 5+)	Transport Layer Security encrypting all HTTP application traffic
HTTPS / HTTP/2	Application-layer protocol carrying API requests and responses
QUIC	UDP-based transport protocol used for HTTP/3 by Meta
ICMP	Network diagnostics — occasional ping/unreachable messages
ARP	Address Resolution Protocol — local MAC-to-IP mapping

4.2 DNS Traffic (Plaintext — High Analytical Value)

DNS queries are unencrypted by default, making them one of the most informative data types. Observed domain resolutions:

- instagram.com, www.instagram.com — primary app domain
- graph.instagram.com — Instagram Graph API endpoint
- i.instagram.com — Instagram image/media CDN
- facebook.com, graph.facebook.com — Facebook main domain and API
- edge-chat.facebook.com — Facebook Messenger/chat service
- scontent.cdninstagram.com — CDN for photos and videos
- static.cdninstagram.com — Static assets (JS, CSS, icons)
- connect.facebook.net — Facebook SDK & analytics
- akamaiedge.net domains — Akamai CDN nodes for media delivery
- Cloudflare infrastructure — CDN and DDoS protection layer

All domains confirmed as Meta-owned or Meta-partnered infrastructure — no unexpected destinations.

4.3 TLS/HTTPS Traffic (Encrypted Application Data)

- TLS version: 1.3 — current industry standard (confirmed via ClientHello/ServerHello)
- Certificate authorities: DigiCert and Let's Encrypt — legitimate, trusted CAs
- Server Name Indication (SNI) visible in TLS handshake — confirms destinations
- No certificate errors or anomalous handshakes observed
- HTTP/2 multiplexing used — multiple streams per connection

4.4 TCP Stream Characteristics

- Standard three-way handshakes (SYN, SYN-ACK, ACK) for all connections
- Connection reuse via HTTP Keep-Alive and HTTP/2 stream multiplexing
- Clean FIN/RST packets upon session termination — no abrupt drops
- No unusual flag combinations (no Xmas scans, no NULL packets)

4.5 QUIC / HTTP/3 Traffic

Facebook connections utilized QUIC, a modern UDP-based transport protocol providing faster 0-RTT establishment, built-in encryption, and reduced latency for mobile networks. QUIC traffic appeared as UDP streams on port 443 — entirely normal for Meta apps.

4.6 ARP & Local Network Traffic

- Standard ARP requests/replies for gateway resolution — normal local network behavior
- No ARP spoofing detected (no conflicting ARP replies for same IP)

5. Security Assessment

5.1 Security Checklist

Security Check	Status	Risk
All HTTP traffic encrypted with TLS 1.3	✓ PASS	None
No plaintext credentials or tokens observed	✓ PASS	None
All certificate authorities are trusted and valid	✓ PASS	None
No connections to unknown or suspicious IPs	✓ PASS	None
No ARP spoofing or poisoning detected	✓ PASS	None
No unexpected outbound ports observed	✓ PASS	None
No malware C2 domain patterns in DNS queries	✓ PASS	None
No data exfiltration volume anomalies	✓ PASS	None
No DNS tunneling patterns detected	✓ PASS	None
All connections terminate to Meta/CDN infrastructure	✓ PASS	None

5.2 Privacy Observations (Informational)

- DNS queries are plaintext — reveal all domains contacted even when content is encrypted. DNS-over-HTTPS (DoH) would prevent this.
- TLS SNI leaks destination hostname in ClientHello — Encrypted Client Hello (ECH) is the emerging solution.
- Traffic timing/volume patterns can reveal behavioral data even without reading encrypted content (traffic analysis).
- QUIC on UDP port 443 is invisible to many basic firewalls configured for TCP only.

6. Key Findings Summary

Finding	Detail
Protocol Types Observed	7+ distinct protocols (DNS, TLS, TCP, UDP, QUIC, ARP, ICMP)

Encryption Standard	TLS 1.3 — highest current standard
Primary Destinations	Meta Platforms, Akamai CDN, Cloudflare
DNS Domains Observed	15+ unique Meta/CDN domains resolved
HTTP Protocol Version	HTTP/2 over TLS; HTTP/3 via QUIC on some streams
Suspicious Activity	None detected
Plaintext Credentials	None found
Overall Security Rating	SECURE — App communicates safely with legitimate infrastructure

7. Skills Demonstrated

Technical Skills

- Wireshark — packet capture, display filters, stream following, protocol dissection
- TCP/IP protocol stack understanding across all OSI layers
- TLS/SSL inspection — handshake analysis, cipher suites, certificate verification
- DNS analysis — query/response reading, domain classification
- Network forensics methodology — systematic evidence review
- Threat assessment — evaluating traffic against known suspicious patterns

Analytical Skills

- Traffic classification and protocol distribution analysis
 - Baseline behavior recognition — distinguishing normal vs anomalous patterns
 - Technical report writing — producing audit-quality findings
 - Privacy-aware analysis — handling captures without exposing personal data
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8. Conclusion

This analysis confirmed that Instagram and Facebook communicate exclusively over encrypted channels (TLS 1.3) with legitimate Meta Platforms infrastructure. No suspicious activity, unauthorized connections, plaintext credentials, or anomalous traffic patterns were identified.

The capture demonstrated a multi-protocol traffic environment including DNS, TCP, TLS, QUIC, ARP, and ICMP — providing a comprehensive real-world dataset. The DNS layer offered meaningful visibility into app behavior even under encryption, highlighting why network-layer monitoring remains essential despite widespread TLS adoption.

Skills developed — packet capture, protocol analysis, certificate verification, DNS inspection, and structured reporting — are directly applicable to SOC analyst, network security engineer, and penetration tester roles.

9. Recommendations & Next Steps

- Enable DNS-over-HTTPS (DoH) on the device to encrypt DNS queries
- Use a VPN to mask IP-level traffic patterns from ISP-level analysis
- Monitor for Encrypted Client Hello (ECH) in future TLS deployments
- Explore certificate pinning bypass using Frida/Objection on a test device
- Future project: compare traffic across multiple social media apps
- Set up Zeek or Suricata IDS to automate anomaly detection on home network

— END OF REPORT —

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